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Sustainable and inclusive energy solutions in refugee camps: Developing a modelling approach for energy demand and alternative renewable power supply

Diana Neves [a,*], Patrícia Baptista [b], João M. Pires [a]

[a] IN+ Centre for Innovation, Technology and Policy Research, Instituto Superior Técnico, Universidade de Lisboa, Portugal
[b] IN+ Centre for Innovation, Technology and Policy Research, Associação para o desenvolvimento do Instituto Superior Técnico, Universidade de Lisboa, Portugal

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ABSTRACT

In long-term emergency operations such as refugee camps, humanitarian aid faces logistical and budgetary limitations. Thus far, the energy systems that supply these operations are designed with little insight on demand needs, leading to the deployment of standardized fossil-dependent solutions which, albeit presenting low investment costs, are pollutant and often inefficient.

This work applies a multi-disciplinary methodology to develop an energy demand modelling framework for Tier 0 refugee camps, and assesses the implementation of alternative power supply systems. It does so by not only focusing on its techno-economic feasibility, but equally through investigating the challenges on adoption in the specific context of refugee camps, by undertaking interviews with relevant stakeholders in the humanitarian sector. The modelling application has been performed for the Mantapala refugee camp, in Zambia, enabling the comparison of different combinations of power supply systems against current diesel-based solutions, using HOMER software.

Results show that the implementation of hybrid renewable energy systems is viable, both technically and economically, with energy costs being reduced up to 50%, with paybacks lower than five years. However, from the interviews performed, some non-technical constraints that may hinder the adoption of these systems in refugee camps were identified, such as uncertainty, lack of funding and difficulties on risk-sharing collaborations. As such, a discussion focused on possible strategies that may help overcome these barriers is presented.

1. Introduction

Humanitarian crises occur anywhere and anytime. Usually politically motivated, these dramatic events can cause severe human and material losses, leaving populations extremely vulnerable. In 2018, 70.8 million people were estimated to be displaced worldwide, 26 million of which were refugees abroad and about half of them, children under 18 years old (UNHCR, 2019a).

Commonly, when a humanitarian crisis occurs, international aid is granted in order to prevent an extension of human and material losses. In 2018, about 81.3 thousand refugees sought resettlement with UNHCR which, depending on geographical location, were settled in refugee camps or within communities (UNHCR, 2019a). Disaster relief operations in refugee camps provide several vital services such as communications, temporary medical clinics, lighting, refrigeration of medicines, water heating and sterilization, all of which require the consumption of energy (Young, 1995). Nevertheless, most refugee camps and large humanitarian operations are commonly set up in regions where access to natural resources, proper infrastructures and power distribution network is already limited. Likewise, host governments are typically reluctant to build permanent and sustainable infrastructures, as humanitarian relief and refugee settlements are usually seen as a temporary situation, albeit currently, the average lifetime of a refugee camp is

Abbreviations: AD, Anaerobic digestion; CFL, Compact Fluorescent Lamp; COE, Cost of Energy; HRES, Hybrid Renewable Energy Systems; LED, Light Emitting Diode; NGO, Non-governmental organizations; NPC, Net Present Cost; PV, Photovoltaic solar energy; UNHCR, United Nations High Commissioner for Refugees.

* Corresponding author. IN+, Instituto Superior Técnico, Av. Rovisco Pais, 1 - 1049-001, Lisboa, Portugal.
E-mail address: diana.neves@tecnico.ulisboa.pt (D. Neves).

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currently approximately 17 years (Global Humanitarian Assistance, 2015).

In these contexts, electricity is predominantly provided by short-term off-grid solutions, such as diesel generators, which guarantee a stable power supply as long as fuel is available. This resource, however, is susceptible to supply shortages, making it hard to comply with the continuous energy needs of humanitarian operations (Fuso Nerini et al., 2015). As noted above, refugee camps can last for decades, raising concerns about their operational budgetary management and environmental impact of relying on fossil fuel-based energy sources (Global Humanitarian Assistance, 2015). Although there is no accurate or systematic reporting of energy data in refugee camps, UNHCR estimates spending a total of 35 million USD every year in diesel fuel to supply electricity (IRENA, 2019). Table 1 presents as example the energy related costs for the Goudoubo (Burkina Faso) and Dadaab (Kenya) refugee camps.

Yet, worldwide, 80% of the 8.7 million refugees and displaced people in camps, have minimal access to energy, with high dependence on traditional biomass for cooking and no access to electricity, with only 11% having access to reliable energy sources for lighting. These numbers demonstrate the urgent need for a more reliable, cleaner, safer and cost-efficient energy access, as a way to help achieve several Sustainable Development Goals (United Nations, 2015), namely the increase of people's well-being and health (by means of clean cooking), the reliability and safety of energy supply (facilitating equality in energy access), and the improvement in education, employment, business and innovation (by providing lighting), among others.

One of the major problems associated with refugee camps regards the activity of cooking. It is estimated that around 85–88% of displaced populations in refugee camps use firewood for cooking (UNHCR, 2019b), which has very low efficiency. This tendency then, goes hand-in-hand with the deforestation of surrounding areas, health risks - due to indoor air pollution (Smith, 2006) contributing to respiratory diseases (Rivoal and Haselip, 2017) -, and ultimately, even a risk to human life - from safety hazards of collecting wood outside camps. In its entirely, cooking related CO_2 emissions in refugee camps are estimated at 5 million tCO_2 per year (Lahn and Grafham, 2015), justifying the need for updated cooking solutions.

However, the lack of reliable and/or metered data on energy usage and consumption in refugee camps' makes it difficult to estimate the real overall energy demand for camp facilities and households, or even the related fuel consumption for electricity supply (Lehne et al., 2016), precipitating the deployment of standardized and frequently unfitting power solutions. In a report by IRENA (2019), diesel generators were reported to fall short of their optimal running load by 25%, leading to increased maintenance costs and fuel losses of around 20%. Moreover, overall inflated costs and emissions are brought about by two mains factors: the absence of harmonizing methods within the energy supply - provided by either the multiple NGOs present in the camps and/or private neighbouring grid suppliers -, and the geographical remoteness of the camps' location - escalating fuel transportation costs (Lahn and Grafham, 2015).

Suffering from lack of funding, as well as short-sighted policies and practices on sustainability and clean energy supply within the humanitarian community, current energy practices in refugee camps are often inefficient, polluting, unsafe for the users and damaging to the surrounding environment. As the number of humanitarian relief operations increase, both in frequency and duration, there is a rising interest in alternative energy systems that could similarly meet energy needs and improve energy access and human comfort (Lahn and Grafham, 2015).

It is estimated that the widespread introduction of improved cookstoves and basic solar lanterns could save $323 million a year in fuel costs in return for a one-time capital investment of $335 million for the equipment. Such an intervention, would also result in emissions' savings of around 6.85 million tCO_2 per year (Lahn and Grafham, 2015).

Researching literature on alternative power supply systems for remote electrification of isolated communities, hybrid renewable energy systems (HRES) are often proposed (Neves et al., 2014). By using local and endogenous resources, HRES can reduce the dependency on fossil fuels, whilst assuring grid resilience and continuous power supply, even when a given energy resource fails or when its price suddenly rises - which is common regarding diesel. If successfully implemented, these systems would not only minimize the environmental impact of power production, but also generate financial savings then available to be channelled towards other crucial humanitarian relief expenses.

Main factors to consider when designing an alternative HRES are the combination of different mature technologies based on the availability of energy resources, fuel prices and distinct consumption profiles (peak/off-peak relation). To comply with these requirements, some studies include diesel generators to work as backup systems, which are only activated when the remaining sources are capable of fulfilling the power consumption (Cerrada and Thomson, 2017; Sigarchian et al., 2015). These can also be adapted to run on biofuels to be produced onsite, using local resources (Salehin et al., 2011; Sigarchian et al., 2015).

However, there is limited research and data reporting on modelling energy demand and alternative supply in refugee camps. Salehin et al. (2011) did a techno-economic analysis of the Energy Emergency Model, consisting on a 20-ft energy supply container that could work on stand-alone or grid-connected mode. It was modelled on a 20,000-refugee camp in Chad. Choosing from a set of different technologies, the authors selected photovoltaics (PV), micro wind turbines, biogas and diesel generators, and batteries as the optimal technology mix for a stand-alone power supply system, reporting a cost of energy (COE) of 0.275 $/kWh. Later on, Fuso Nerini et al. (2015) continued exploring that same concept for Ouagadougou, a 5000-people settlement in Burkina Faso, considering a combination of PV, wind and a dual-fuel generator (biomass and gasoline). The authors concluded that such a system could fulfil the basic needs of the camp management, in a cost competitive way, with a COE of 0.4 $/kWh. However, the

Table 1
Energy spending costs on typical refugee camps (Lahn and Grafham, 2015).

	Goudoubo (Burkina Faso)	Dadaab (Kenya)
Population	10,327	356,014
Costs with diesel fuel for power operations	31.5 k$/year	2.3 M$/year
Energy related costs on global budget[a]	8%	20%
Average monthly energy spending per household[b]	10.6 $/month (5–7% of total income)	17.20 $/month (24% of total income)

[a] Incorporating costs related to camp facilities such as schools, hospitals, refugee community facilities, etc., infrastructure items such as water pumps, filters, streetlights, etc. and transportation of goods and persons.
[b] Regarding lighting and cooking.

demand profile used was modelled on a minimal set of administrative energy end-uses of humanitarian disaster relief, such as water purification, lighting of common spaces, vaccines and telecommunications, which ultimately reveals that only a limited part of the base camp energy needs were being fulfilled.

Cerrada and Thomson (2017) have explored different business models regarding the adoption of PV microgrids in refugee camps. Considering a 5000-refugee camp in Bahn (Liberia), they modelled three supply scenarios: providing lighting and mobile phone charging to refugees (PV and batteries); providing administrative end-uses for camp management, such as lighting, health and administration services; and a combination of the two scenarios (considering PV, batteries and diesel generator). Their results demonstrated the viability of selling electricity to refugee inhabitants at a rate of 1 $/kWh, given a diesel price higher that 0.6 $/l. The profit obtained would increase when also providing electricity for camp management operations. Nonetheless, this last solution presents higher risks, as it requires higher initial investment and a considerable sensitivity to diesel price, which is a key factor in refugee camps.

Chowdhury et al. (2020), formulated to the Kutupalong refugee camp in Bangladesh, a stand-alone system consisting of PV, wind, batteries and diesel to supply lighting, cooling, water pumping and electronic devices' needs for a hypothetical Rohingya community, yet not mentioning the scale of the community (number of refugees or households). Results point for a COE of 0.35 $/kWh and CO_2 emissions reduction of 84% of regarding an only-diesel system.

Ossenbrink et al. (2018) studied the deployment of PV for water pumping, finding a levelized cost of energy (LCOE) from 0.1 to 0.3 $/kWh for three different refugee camps: Dadaab (Kenya), Mafraq Za'atari (Jordan) and Harran-Kokenli (Turkey). Furthermore, they show that PV-grids would be a viable option for 70% of the largest UNHCR refugee camps which detain a solar global horizontal irradiation higher than 2000 kWh/m^2year, which however remain untapped.

Subsequently, in the analysed literature, regarding the energy demand profiles utilised, we conclude that each case study presents a unique case-based energy demand modelling, exposing a limitation when trying to systematize the rationale behind the amount of the chosen loads. Further, as no information is disclosed in terms of indicators per refugee or per household, we witness a knowledge gap in the building of a systematized approach to the optimal (or adequate) level of energy service associated to each energy use in refugee camps.

Nonetheless, there are pilot projects that albeit not approaching the energy needs of refugee camps as a whole, try to give response to specific refugee energy needs on the settlements. Such an example is the campaign *Brighter Future for Refugees* (UNHCR & IKEA Foundation, 2015), focusing on distributing solar lanterns to camp inhabitants for lighting purposes (for their shelters or mobile lighting around the camp) and phone charging, or the installation of solar-powered street lights, supporting increased security and allowing for communal activities after nightfall. These kinds of initiatives have visible positive social impacts namely on education, business, and refugees' security. Another alternative that seeks to make access to water ever cleaner and more sustainable, is the installation solar PV water pumping systems. Otherwise, water would need to be transported or pumped using diesel as primary source (IRENA, 2019). Further, the installation of PV-power plants can also serve other end-uses, and communities (as the one installed in Azraq camp, Jordan). Likewise, these campaigns intend to innovate, upgrading to cleaner cooking solutions by collecting the biowaste from latrines for use in communal kitchens (UNHCR & IKEA Foundation, 2015; Eyrard et al., 2015), and thus solving the waste management of the Kutupalong camp in Bangladesh. Similarly, other pilot solutions have been adopted, so diverse as to using bioenergy for clean cooking with biomass briquettes (IRENA, 2018; United Nations Industrial Development Organization, 2016), deploying PV-battery home systems (IRENA, 2019) or grid connected PV power plants (KISEDP, 2016).

Lastly, despite being apparently straightforward, the adoption of such systems in refugee camps can face several barriers, which go far beyond the technical spectrum, often involving the management and operational culture of humanitarian organizations. Such constraints imply that any attempt to change practices within humanitarian relief operations should not only focus on technical and economic factors, but also consider the social and operational variables of this specific ecosystem.

As a result, in order to pave the way for a sustainable transition and identify the most cost-effective, low-carbon energy systems' alternative for refugee camps, this work aims to respond to the identified literature gaps on modelling energy demand and deploying renewable energy systems in refugee camps by:

- developing an energy demand modelling framework that identifies typical energy end-uses and needs on refugee camps considering the World Bank's (World bank, 2015) various Tiers of energy access, thus resulting in more accurate demand profiles according to the type of refugee settlement architecture;
- conducting multiple interviews with stakeholders from the humanitarian sector, with a focus on identifying different opportunities and barriers in the adoption of alternative power supply systems;
- performing a techno-economic analysis of alternative renewable power and cooking supply technologies, assessed in terms of costs, energy demand and CO_2 emissions, as compared with current diesel-based solutions; and
- analysing and discussing the proposed solutions in light of the existing literature and influencing factors, namely by suggesting a possible business model intended at overcoming the barriers to adoption.

For the purpose of this analysis, the refugee camp of Mantapala, in Zambia, was chosen as a case-study. By applying a multi-disciplinary methodology, focusing not only on the technical and economic challenges of implementing alternative power supply systems, but also through bringing a human perspective by accounting for social and operational variables found in refugee camps, often neglected in previous studies, this study enables a holistic approach to energy provision in a safe and sustainable way, singularly in such a complex context as refugee camps.

The paper is organized as follows: Section 2 presents the methodology for assessing energy demand and alternative power supply systems in refugee camps, Section 3 presents the case-study details and modelling approach, Section 4 reports the results of both the interviews and the energy demand and energy supply modelling, while Section 5 discusses the findings. Lastly, in Section 6, the final statements are presented.

2. Methodology

Given the identified literature gap, this study comprehends two goals: designing a modelling framework for assessing energy demand in refugee camps, based on an energy indicator per household; and assessing different, more affordable, resilient and sustainable power supply solutions for refugee camps. Thus, a multi-disciplinary methodology was applied, combining inputs from distinct case studies, electricity access' Tiers, interviews with relevant stakeholders and results from power systems' simulations. Fig. 1 presents the layout of the adopted approach.

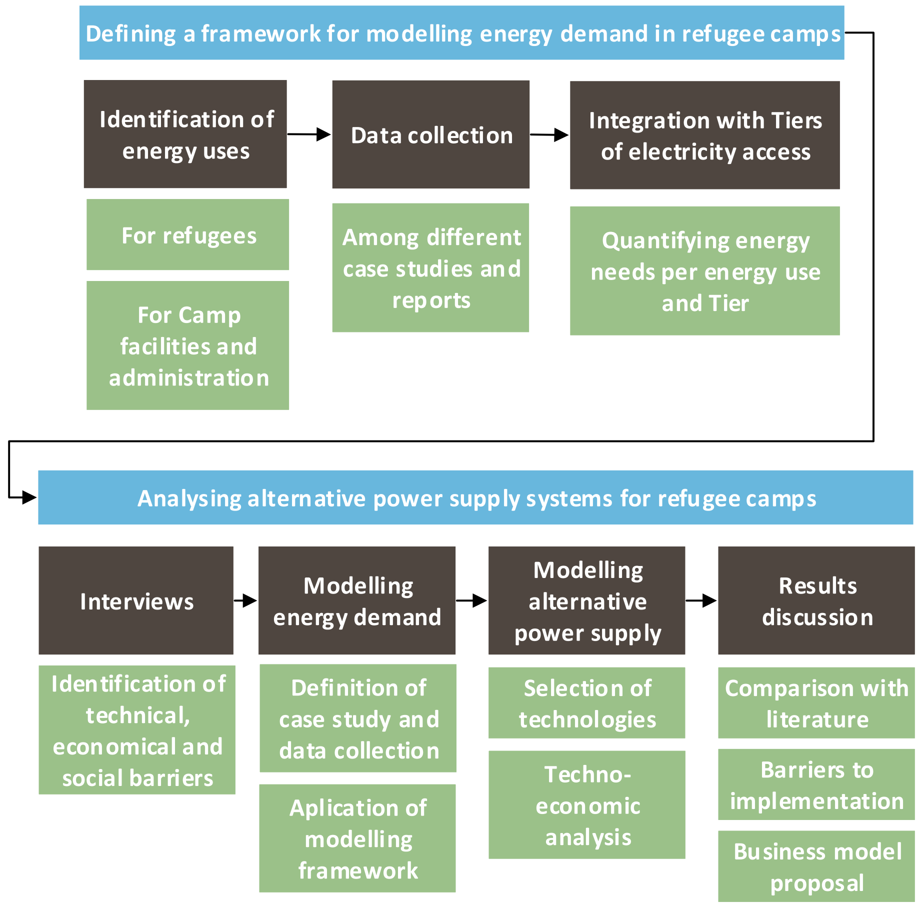

Fig. 1. Methodology flowchart.

2.1. Modelling energy demand in refugee camps

As previously seen, the literature suggests that diesel generators are responsible for the majority of electricity delivered to refugee camps, as well as biomass burning for cooking purposes, leading to increased pollutant emissions and harmful environmental and health impacts. In May 2014, the UNHCR launched its first energy strategy, established on the idea that all refugees '*should be able to satisfy their fuel and energy needs for cooking, heating, and lighting in a safe and sustainable manner, without fear or risk to their health, well-being, and personal security*'. However, with minimal energy data and/or related expertise within refugee camps, assessing accurate estimations of energy demand to analyse the best possible alternative energy systems, becomes of crucial importance. Given the lack of specific reference to heating needs, only electricity and cooking needs will be explored, assuming that part of the heating needs are assured by cooking (Lehne et al., 2016).

2.1.1. Assessing electricity needs

One of the reasons for the limited existing knowledge on refugee camps' electricity needs, is the low rate of electricity access found among settlements' households, forcing informal connections and access to neighbouring grids or diesel generators, most often not metered or accounted for in anyway. Hence, for assessing the households' needs, we will be using the World Bank's categorization for households' energy access, which is divided in six Tiers, as presented in Table 2. For each Tier, indicative electric appliances, their minimum power and availability are detailed in Table 3, as well as the minimum daily energy demand and typical energy system solutions.

According to Lahn and Grafham (2015), 89% of people in refugee camps have Tier 0 lighting, while an estimated 80% have Tier 0 cooking facilities. Consequently, so as to provide insights about energy demand on off-grid solutions, our modelling work will be focusing on Tier 0 to Tier 3, which englobes the majority of refugee camps, given that for Tier 4 and 5 settlements, most of the solutions are grid connected, where metering technologies can be applied.

Regarding energy needs of the camp infrastructure and management, we normally observe an operations/office centre, staff accommodations and health centre. Depending on the size of the camp, refugees' age and average time of stay, schools and community centres (for training or other livelihood programs), might be also envisioned. Street lighting and other basic services such as water access are also considered as part of camp infrastructure. Table 4 reports some requirements for planning camp facilities, whilst Table 5 presents the estimated energy and power loads for the camp's infrastructures, as detailed as possible. However, these would need to be adapted according to the size of the settlement considered, in order to provide an electricity demand modelling basis.

Hence, the daily electricity needs per household ($E_{hh,elect}$) as well as for infrastructure & camp facilities ($E_{camp\ infr\&facil,elect}$) would be calculated by the sum of the electricity demand of the aforementioned end-uses, and adapted for each camp in analysis. Considering five as the average number of refugees per household (UNHCR, 2016), we can find the number of households per camp

Table 2
Definition of electricity access tiers (Source: adapted from (Lehne et al., 2016), (World bank, 2015)).

		Tier 0	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Indicative Appliances		None, kerosene lamps or candles	Task lighting + phone charging/radio	Tier 1 + General lighting + television + fan (possibly)	Tier 2 + medium-power appliances	Tier 3 + high-power or continuous appliances	Tier 4 + Heavy-power appliances
Minimum power capacity ratings	**Power [W]**	No electricity	3	50	200	800	2000
	Energy [Wh]		12	200	1000	3400	8200
Availability	**Hours per day [h]**	<4	4–8	4–8	8–16	16–22	>22
	Hours per evening [h]	<1	1–2	2–3	3–4	>4	>4
Annual demand	**[kWh]**		≥4.5	≥73	≥365	≥1250	≥3000
Daily demand	**[Wh]**		≥12	≥200	≥1000	≥3425	≥8219
Affordability		Not affordable		Affordable (<5% of household income)			
Reliability		Unreliable				Mostly Reliable	
Quality		Poor quality of electricity supply			Good quality of electricity supply		
Legality of supply		Illegal energy supply			Legal energy supply		
Health and safeness		Unhealthy and unsafe energy system				Healthy and safe energy system	

Table 3
Household end-uses detail.

					Tier 0	Tier 1		Tier 2		Tier 3		
Household End-uses	**Equipment**	**Power [W]**	**Usage [h/day]**	**Charging/usage schedule**		**Energy [Wh/day]**	**Number per household**	**Energy [Wh/day]**	**Number per household**	**Energy [Wh/day]**	**Number per household**	**Adapted from sources:**
Lighting	LEDs	3	4	6pm–10pm	Candles or kerosene lamps	12	1	60	5	60	5	Cerrada (2016)
Small electric appliances	Mobile phone charging[a]	3	4	8am-6pm (during solar hours)		12	1; charged every day	12	1; charged every day	0	2; charged every day	Cerrada (2016)
	Fan	15	8	10am-4pm				120	1	120	1	Fuso Nerini et al. (2015)
	Television	100	2	8pm–10pm				200	1	200	1	
	Radio	5	4	9am–11am; 5pm–7pm				20	1	20	1	
Medium-power appliances (considered communal)	Washing machine	1500	1 cycle per day	10am–12am						3000	1 per 100 people; or 1 per 20 households	Norwegian Refugee Council (2008)
	Electric kettle	2000	0.5	10 min at 8am; 10 min at 12am; 10 min at 7pm						1000	1 per 10 households	
Total Energy [Wh]						24		412		4400		
Energy system typically proposed					Solar lighting kits	Stand-alone or home solar systems		Generator or Mini-grid		Generator or grid		IRENA (2019)

[a] For nominal simultaneous power, it is considered that, in maximum,1/3 of the total amount of mobile phones in the camp charge simultaneously.

Table 4
Camp management facilities (Adapted from sources (Cerrada, 2016;,Norwegian Refugee Council, 2008; UNHCR, 2016):).

Description	Standard
Water supply	20 L per person per day
Health centre	1 per 20,000 persons
Referral hospital	1 per 200,000 persons
School	1 per 5000 persons; 1 teacher for 40 pupils
Market place	1 per 20,000 persons
Feeding centre	1 per 20,000 persons
Lighting	As appropriate
Administration/office	As appropriate
Staff member	1 per 200 refugees

($N_{households}$) and the overall electricity needs of a settlement by Equation (1).

$$\begin{cases} N_{households} = \dfrac{N_{camp\ population}}{5} \\ E_{settlement\ ,elect} = \left(N_{households} \times E_{hh,elect}\right) + E_{camp\ infr\&facil,elect} \end{cases} \quad (1)$$

2.1.2. Assessing cooking needs

As previously seen, firewood is the primary source of energy used for cooking in refugee camps, and extremely time consuming: refugees report spending 19–20 h per week collecting firewood (UNHCR, 2019b), plus 6 h per day on the activity of cooking (Rivoal and Haselip, 2017), hindering their capacity to attend school or earn an income.

It is therefore important to quantify the energy need associated

Table 5
Camp infrastructure equipment energy consumption.

Camp Infrastructures	Equipment	Description	Power	Usage [h/day]	Charging/usage schedule	Energy [Wh/day]	Assumptions	Adapted from
Internal lighting	CFL	15 W CFL bulbs	15 W	5	6pm–11pm	75	Natural lighting is considered during the day, and internal lighting is only considered for staff accommodations. The unitary value should be multiplied by constructed area (7m^2 per staff member), considering 1 CFL bulb per 5m^2	Cerrada (2016)
Staff Accommodation	Small power (phone/tablet charging)	6 W mobile phone; 12 W tablet	18 W	4	10pm to 2am	72	1 per staff member; charged every day	(Harkness et al., 2017; UNEP/DFS/UNSOA, 2010)
Public/security Lighting	LED	20 W LED units	20 W	10	6pm-4am	200	Public lighting should be provided to light streets and paths within the households' area, toilets, and outside the common spaces of the camp's central infrastructures. Thus, it is assumed that: - 1 lamp should be installed outside each household to provide streetlight, without the need of installation poles; - 1 lamp per 4 latrines should be contemplated, considering that each latrine serves 20 refugees; - 15 lamps should be installed in the common area of the camp management (health centre, schools, administration, etc).	Cerrada (2016)
Water access	Water pumping	37 kW typical pump size for 240,000 l/day	37 kW	7	9am-4pm (peak solar hours assumed)	259,000	Assuming 20 l/refugee per day, this system would allow providing water for 12,000 refugees; this load should be adapted to camp population according to instructions in (IOM et al., 2018)	(IOM et al., 2018)
	Water Purification	Unit of reverse osmosis	2.25 kW	24	8am-9pm	54,000	In most camps, water purification is done through chlorination (Safe Drinking Water Foundation, 2020); however, Fuso Nerini et al. report a water purification unit that could be used: This unit provides for 5000 l/day, working for 13 h h. If working 24h, could assure 9231 l/day. Assuming 20 l/refugee per day, this value should be adapted according to the needs of each camp.	Fuso Nerini et al. (2015)
Operations management	Wi-Fi router		15 W	24	0am-12 pm	360		Cerrada (2016)
	Satellite Telephone		60 W	12	9am-9pm	720		Cerrada (2016)
	Mobile phone charging		3 W	4	9am-5pm	12	Unit value; 1 mobile per staff member; for nominal power in the camp, consider 1/3 charging simultaneously	Cerrada (2016)
	Computers	Laptop	85 W	6	10am-2pm (charging 4h during solar hours)	340	Unit value; Considering a computer ratio of 3:1 staff member, and that only 50% of accounted staff members work on operational managing offices	Harkness et al. (2017)
	Printer		40 W	8	9am-5pm	320	1 printer	Cerrada (2016)
	Climatization	Fans	75 W	8	9am-5pm	600	Although some references consider natural ventilation (UNEP/DFS/UNSOA, 2010), it is important to consider some kind of climatization. Thus, a ceiling fan is considered	Shah et al. (2015)
School/Community centre	Desktop Computer + Projector		409 W	7	9am-4pm	2863	Most of the schools are only equipped with chalkboards and consider natural lighting. However, for basic needs, 1 set of computer + projector per school and one 20 W LED lamp per classroom was considered to fulfil school and community programs purposes; Though, according to specific sponsorship programmes (UNHCR, 2018a), the number of electric equipment can differ.	(Harkness et al., 2017; EPSON, n.d.)
	Internal lighting	LED	20 W	7	9am-4pm	140		
Health care facilities	Internal lighting + health equipment	Vaccines cooler, refrigerator, incubator, fan, other health equipment	2 kW[a]	24	0am-12 pm	10,000	Value assumed for a small clinic with capacity for 5 beds (for a 5000 refugee camp); the equipment considered is detailed in (Cerrada, 2016)	Cerrada (2016)

[a] nominal simultaneous power.

to cooking. Taking into consideration the average number of people per household - normally 5 persons (Lahn and Grafham, 2015) - and that each household is responsible for the consumption of 17.5 kg of firewood per day (IRENA, 2019), it is possible to estimate the firewood demand per camp. Taking into consideration an average firewood calorific value of 13.3 MJ/kg (AEBIOM, 2009) and a burning efficiency of 15%, in this case (WLPGA, 2018), it is possible to quantify the daily useful energy ($E_{u,cooking}$) required for the activity of cooking, as presented in Equation (2).

$$E_{u,cooking}\left[\frac{MJ}{day}\right] = N_{households} \times E_{firewood}\left[\frac{kg}{(day.household)}\right] \times CV_{firewood}\left[\frac{MJ}{kg}\right] \times \eta_{firewood} \tag{2}$$

where $N_{households}$ is the number of households in the camp, $E_{firewood}$ is the firewood need per household per day, $CV_{firewood}$ is the calorific value of firewood, and $\eta_{firewood}$ is the efficiency of burning firewood. Further, the associated cooking CO_2 emissions can be obtained by Equation (3):

$$CO_{2\,emi,cooking}\left[\frac{kg_{CO2eq}}{day}\right] = E_{fuel} \times \varepsilon_{fuel} \times N_{households} \tag{3}$$

where E_{fuel} is the consumed fuel for cooking per household per day and ε_{fuel} is the fuel's CO_2 emission factor.

2.2. Analysing alternative power supply systems

2.2.1. Interviews

Studies usually focus on displaying technical and economical results for a determined camp site, however neglecting to identify and discuss, in parallel, potential challenges associated with the specific, practical adoption in real world and outside each case-study. Consequently, this was the principal motivation when interviewing various relevant stakeholders in the field.

To analyse the viability and acceptance of alternative power supply systems, 8 semi-structured in-depth interviews were conducted with professionals from various backgrounds and organizations. The sample ranged from researchers focused on innovating energy and health systems amidst humanitarian emergency scenarios (TU Delft and NTNU), and on hybrid renewable energy systems (KTH); collaborators from Non-Governmental Organizations (NGO) with energy efficiency projects in rural African villages (Inyenyeri and TechnoServe); health professionals from an international medical assistance emergency provider (AMI); engineers working on power supply systems, and social entrepreneurs. At the time of the interviews, all of them were working on projects related with humanitarian emergency operations and/or electrification of isolated communities.

The interviews were structured to cover two main topics. First, to enable the characterization of the humanitarian ecosystem providing inputs for techno-economic analysis and, secondly, to identify the most critical barriers to the introduction of alternative power supply systems in refugee camps, making the evaluation of different options better justified.

2.2.2. Modelling software and suitable renewable technologies

For modelling the alternative power supply system, HOMER software (HOMER Energy, 2010) which is a modelling and planning tool, was chosen. It evaluates the optimized solution for different technical and economical scenarios, while providing detailed technical and economic parameters for all possible energy systems' configurations. Its optimization is performed based on the Net Present Cost (NPC), considering technical constraints, returning as outputs other economical parameters as the payback and cost of energy (COE) of a certain solution.

Based on the literature review, to perform a techno-economic analysis, three distinct technologies emerge as possible alternatives to the current only-diesel generators: photovoltaic panels (PV), micro wind turbines, biogas generators.

PV and micro wind turbines are also commonly suggested in the literature as viable technologies for stand-alone power systems. Their presence and share on these systems depend strongly on solar radiation and wind available on site. However, due to the variability of weather conditions, both technologies can complement each other, minimizing the risk of power outages. Financially, PV and micro wind turbines present higher acquisition costs than diesel generators, although being extremely inexpensive to operate since they do not require fuel expenditures. Moreover, as both technologies start reaching their maturity stage, significant price reductions have been observed, especially for PV systems, and are expected in the next few years (Fraunhofer ISE, 2015). This can increase interest in introducing these solutions on stand-alone power systems, as the investment becomes less risky and easier to payback.

Due to the intermittent power production output of some technologies and to secure against consumption peaks, energy storage units can be a solid addition to a power system. As concluded by Nikolaidis and Poullikkas (2017), lithium batteries currently offer better technical specifications (e.g. higher efficiency, energy density and lifetime, rapid charge and discharge capabilities), but lead-acid batteries are still the most cost-effective solution, mainly due to low upfront costs and acceptable performance in harsh weather conditions, such as the ones found in camps. Therefore, the latter are typically more likely to be proposed.

The use of biogas is also frequently suggested in the literature as an alternative electricity supply in remote locations (Sigarchian et al., 2015). In fact, as will be seen in section 2.2.3., the use of locally produced biogas can solve three major problems currently found in refugee camps: cooking energy needs, power supply, and waste management.

It is also worth mentioning that, although this work focuses on finding alternatives to diesel generators, the latter technology should not be immediately excluded from the modelled system. Its reliability for base load generation, low investment cost and technologic maturity makes diesel generators a valid choice, especially as a backup system during both solar radiation and wind shortages. The goal is to minimize the dependency on diesel generators and not necessarily remove them from the power supply system, since the majority of the settlements have them already deployed.

2.2.3. Biogas production potential

In this work, employing biogas is proposed as an alternative energy source for cooking. By collecting and processing the daily human and organic waste produced inside refugee camps, a considerable volume of biogas can be obtained and used either for cooking or for power production. In parallel, the implementation of a waste disposal and management system might allow humanitarian organizations to suppress some of the public health problems found inside refugee camps (Mahamud et al., 2012). Although technically, the production of biogas from such resources is considered simple, from a social perspective this process can be more complex. Aside from the fact that several cultures are highly reluctant to handle or reuse human waste, the large population and dimension of refugee camps creates logistical challenges for the local production of biogas. These social factors have usually been disregarded or even neglected in the previous studies. Salehin et al. (2011), however, proposed introducing educational initiatives to promote human waste collection, thus minimizing social rejection surrounding this practice.

Considering a 58% efficiency in the cooking process using biogas (Santopaolo, 2019), the final energy ($E_{f,biogas}$) required of biogas can be estimated per day, as presented in Equation (4). This corresponds to the energy demand of biogas that must be produced to fulfil the cooking needs of the camp. This can be converted to volume (m^3) by considering a calorific content of 22 MJ/m^3 (Roubík et al., 2018).

$$E_{f,biogas}\left[\frac{MJ}{day}\right] = \frac{E_{u,cooking}\left[\frac{MJ}{day}\right]}{\eta_{biogas}} \tag{4}$$

Considering the production of biogas using human waste as a resource, and counting the total population of refugees in a camp, it is possible to estimate the waste generation in latrines, suitable for biogas production. Taking into account the daily human feces generation (in this case, considering an average value of 0.19 kg/day from references (Andriani et al., 2015; Mitiku Teferra and Wubu, 2019; Santopaolo, 2019)) and a percentage of volatile solids[1] (VS) of 21% (average value based on (Andriani et al., 2015; Mitiku Teferra and Wubu, 2019; Santopaolo, 2019)), it is possible to scale and estimate the quantity of volatile solids produced by the entire refugee population. Assuming that not all waste is collected (Salehin et al., 2011), a collection factor was considered, to guarantee that the assumptions are consistent with the conditions present in refugee camps. Consequently, a 50% collection factor (CF) was considered for human feces (Salehin et al., 2011).

The selected production process for biogas production is anaerobic digestion (AD), due to its capacity to break down organic waste in the absence of air, and the added advantage of producing a secondary product (the digest) that can be used for agricultural purposes, as fertilizer (Salehin et al., 2011).

Regarding biogas productivity, the literature indicates that the yield of biogas production from human feces is low, as a result of the low carbon to nitrogen (C/N) ratio, which may require supplementing the inflow with grass, sawdust or paper, in order to improve the biogas production yield (Ismail and Adewole, 2014). In this case, an average value of 0.22 $m^3/kg_{volatile\ solids}$ from references (Mitiku Teferra and Wubu, 2019; Regattieri et al., 2018; Santopaolo, 2019) was considered. Additionally, since it is expected that the biogas operational conditions may not be entirely optimized, a safety margin (SM) was considered: in this case 50% (Salehin et al., 2011). Another component that must be considered is that the biogas production process through AD, typically, has an energy consumption ($EC_{biogas\ prod.}$) of around 15% of the biogas produced (Labriet et al., 2013). In this manner, the biogas production potential can be expressed by Equation (5).

$$\begin{aligned} E_{biogas}\left[\frac{MJ}{day}\right] &= N_{refugees} \times feedstock_{biogas} \times \%VS \times CF \\ &\quad \times \text{biogas production yield}\left[\frac{m^3}{kg_{VS}}\right] \times SM \\ &\quad \times CV_{biogas}\left[\frac{MJ}{m^3}\right] \times \left(1 - EC_{biogas\ prod.}\right) \end{aligned} \tag{5}$$

where $N_{refugees}$, is the number of refugees, $feedstock_{biogas}$ is the amount of daily human feces generated, VS the percentage of volatile solids, SM the safety margin for the operation of the biogas system, CV_{biogas} the calorific value of biogas, and $EC_{biogas\ prod.}$ the energy consumption of the anaerobic digester.

It should be stated that one of the pre-requisite for building a biogas plant is having access to water, so this solution would be avoided in water scarce locations (Energypedia, 2018).

By comparing the production potential of biogas per day (MJ/day) (Equation (5)), with the daily final energy demand of biogas for cooking (Equation (4)), it is possible to evaluate how large the contribution of biogas production can be either to cooking or to electricity production.

Nonetheless, behavioural change within these populations can become complicated and time-consuming. An example is the variety of past unsuccessful initiatives to promote energy efficient cook stoves in refugee camps (The World Bank, 2011). Although significantly reducing the prevalence of respiratory diseases and allowing considerable fuel savings for refugees, most improved cook stoves have very low adoption rates. Researchers found that these solutions often failed because they demanded the adoption of new cooking practices. Either by having to stand up to check the pot or simply because they missed the previous smoky flavour of the food, refugees rapidly returned to their old cooking apparatus.

Alternatively, instead of changing the cooking habits of the entire population, it could be possible to feed biogas directly into a gas-fired combustion turbine to produce electricity (Andriani et al., 2015).

Therefore, a small taskforce of refugees would be created and trained to collect human waste from across the camp and store it, to be later transformed into biogas. If successfully implemented, this strategy could finance the waste management process with savings from using biogas as fuel.

3. Case-study

3.1. Mantapala Refugee camp

The Mantapala Refugee Camp (9°30′58.824″S, 28°53′13.695″E), in the Nchelenge district, in northern Zambia, was setup in early 2018 to host refugees mainly from the Democratic Republic of Congo. Originally planned to host around 20,000 refugees (UNHCR, 2018b), in 2020 it hosted 14,319 refugees (UNHCR, 2020), the majority farmers, traders or artisans. This settlement is considered to have a new approach to refugees' movements across the world, since it is expected to improve the local development and thus livelihood, by providing and developing farming and trading opportunities for both hosts and refugees (UNHCR, 2018b; UNHCR, 2018c). With an allocation of 0.5 ha for farming and the deployment of three markets to allow trading between the community and the settlement, it has additionally elected a voluntary community leadership to foster the community interest in decision making (UNHCR, 2019c). According to Jahre et al. (2018), which proposes and systematizes design approaches to refugee camps, the settlement is aligned with the new ideal refugee camp design, empowering both the refugees and community with the goal of building longer-term development and autonomous livelihood.

The Mantapala refugee camp was chosen as case-study since it is on an early-expansion phase, still with potential flexibility to adapt its structure, opening up opportunities for the deployment of alternative power supply systems (which can be more easily installed than in older and larger refugee camps), benefiting not only the refugee camp but also the surrounding community. Moreover, its location on the sub-Saharan African context provides the possibility of replication to other camps, or even for small communities since in the rest of the country, 70% of households have electricity access equal or lower than Tier 3, with 60% having Tier 0 (Mini-grids partnership et al., 2020).

[1] The percentage of volatile solids corresponds to the fraction of total solids which is composed mainly of organic matter. They correspond to the quantity of total solids that are lost when the residues are combusted at 550 °C, in excess air.

Table 6
Mantapala refugee camp facilities (source (UNHCR, 2020) (UNHCR, 2019c) (UNHCR, 2019d) (Filipová, 2018):).

Energy consuming camp facilities	Description
Households	around 4245 households, with average occupancy of 4 persons; plots with 30 × 25m; rely on firewood, candles, solar panels and batteries for lighting and domestic use
School	2 schools, total 29 classrooms, with 6822 enrolled pupils, with overpopulated classrooms (1:74 teacher/learner ratio); two new school blocks are being constructed, totalizing 35 classrooms, targeting 1:52 ratio (against the standard 1:45)
Health centre	temporary clinic provides basic services, while a permanent one is under construction
Water access	55 boreholes, supplying 423 m^3/day or 28.4 l/day.refugee; 2 motorized water systems and numerous hand pumps; chlorinated treatment is done every two weeks for water purification purposes (World Vision International, 2020)
Waste management	52% of the households (2203) have latrines, while the remaining 48% use temporary ones
Staff Accommodation	no staff accommodation has been reported
Offices	Established in temporary *Better Shelter* solutions (Better Shelter, 2019)

Table 6 presents the energy consuming activities seen in Mantapala refugee settlement, which, as previously reported, has the particularity of allocating small areas of land to each household for subsistence farming.

In the Nchelengue district, where Mantapala Camp is settled, less than 20% of the population has access to electricity, using mostly charcoal and firewood for cooking, even among the few households with electricity. Mantapala Refugee camp is not connected to the electricity grid (and is not expected to be in the near future), being supplied by diesel generators and individual solar energy initiatives, namely by the 10 public lighting poles installed and the solar lighting kits (Filipová, 2018).

Due to lack of on-site data regarding Mantapala's energy demand, the modelling framework developed in Section 2.1 was implemented, estimating the desirable power consumption profile of the entire settlement, presented in Section 4.1.

3.2. Simulation model description

Based on the technological choices presented in section 2.2.2, the stand-alone power production system was modelled in HOMER, as shown in Fig. 2, with its techno-economic characteristics presented in Table 7.

PV panels, wind turbine and battery bank were conceptualised as connected to a DC-Bus, while diesel and biogas generators are connected to an AC-Bus. These two buses are both connected through a bi-directional inverter, which is also included in the model simulation. Additionally, the load demand was assumed to be AC.

The model was allowed to consider any combination of the previous technologies, within a predefined range for each technology (as presented in Table 7), as long as they were able to continuously fulfil the power demand of the refugee camp. The baseline scenario was considered to be a diesel-only powered system to which all other possible systems are compared. For the case studied, the diesel's acquisition cost was assumed as 0.87 $/l (Global Petrol Prices, 2020), while the solar and wind potential are presented in the Annex (Fig. 5 and 6)).

The simulations performed were compared in terms of initial investment, cost of energy, payback time and CO_2 emissions, allowing the possibility to identify the most appropriate solutions according to the different criteria.

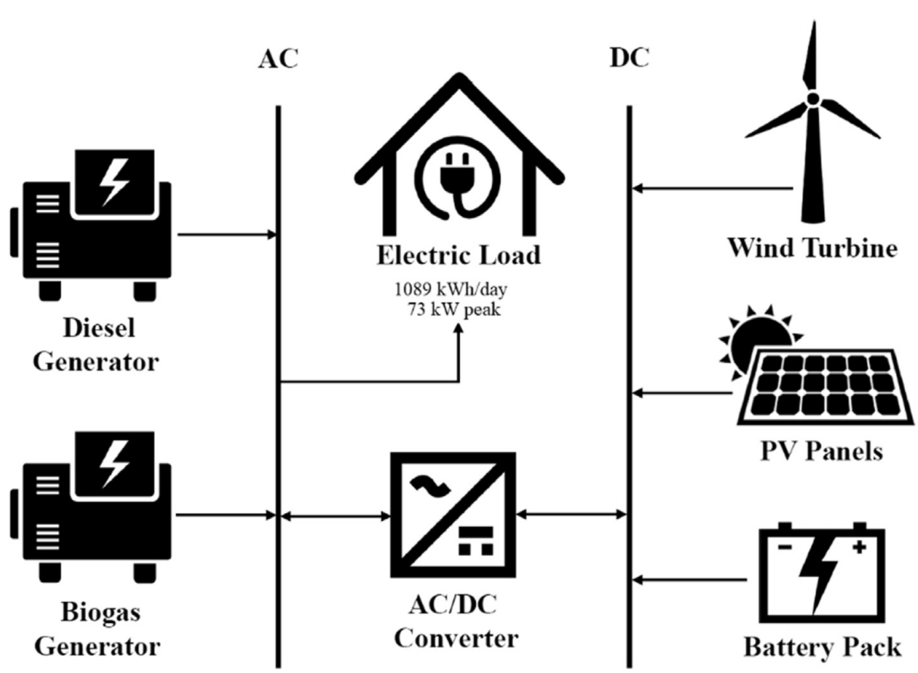

Fig. 2. - Schematic of the stand-alone power supply modelled for the simulations performed in HOMER software.

Table 7
Techno-economical characteristics of selected technologies (Source (Moner-Girona et al., 2019; NREL, 2016; Sigarchian et al., 2015; Mandal et al., 2017).

Technology	Cost [$/kW]	Replacement Cost [$/kW]	O&M Cost	Lifetime	Ranges of installed capacity considered [kW]
Diesel generators	600	500	0.02 $/hr	15,000 running hours	Autosized
Photovoltaic panels	933	933	19 $/kW.yr	20 years	0–500
Micro Wind turbines	2022	2022	40 $/kW.yr	14 years	0–100
Biogas generators	1500	1200	0.10 $/hr	15,000 running hours	0;20;40;60,80;100;120
Batteries	393 $/kWh	393 $/kWh	15 $/yr	10 years	0–500

For the base-case scenario, the duration of the project was assumed to be 10 years. It was also considered that governmental entities would finance the system, therefore resulting in a 0% real discount rate for the investment. Although conservative, this value aims to be more realistic regarding the duration of humanitarian relief operations, as most organizations tend not to stay for the entire lifetime of refugee camps, due to donor fatigue.

4. Results

4.1. Modelling energy demand and biogas production potential for Mantapala refugee camp

In the absence of specific energy consumption data, in this section, the Mantapala refugee camp energy demand is modelled applying the methodology developed in section 2.1, for both the electricity and cooking needs.

4.1.1. Electricity needs

For the electricity needs, both individual household needs and the power required for camp management operations, were estimated according to planning standards. Fig. 3 presents the modelled power load profile, while a table with the calculations' corresponding detail and assumptions is presented in the Annex (Table 11).

An overall daily demand of 1089 kWh and a peak of 73 kW are estimated. However, to account for fluctuations of power consumption within the refugee camp, a daily and timestep-to-timestep variability factor of 10% were imposed on the model, leading to a peak of around 107 kW. Moreover, for better visualization, the buildings' electricity needs were separated from the public services' needs - water pumping and public lighting - since these are the sectors with most associated uncertainty, given there is no detailed information available about the head of the pumps or if all the water demand (28.4 l/day. refugee) is purified or not.

Nevertheless, a critic observation of this methodology's implementation of should be made. As reported in Section 1, 89% of the camps have Tier 0 electricity access, namely on lighting access, meaning that lighting is inexistent or assured by candles or kerosene lamps. In this study, we tried to model the minimum standards of lighting and electricity access, which means that Tier 0 situations would be at least improved to Tier 1. Correspondingly, this results in higher modelled electricity demand than the currently existing in refugee camps, since it is expectable that with

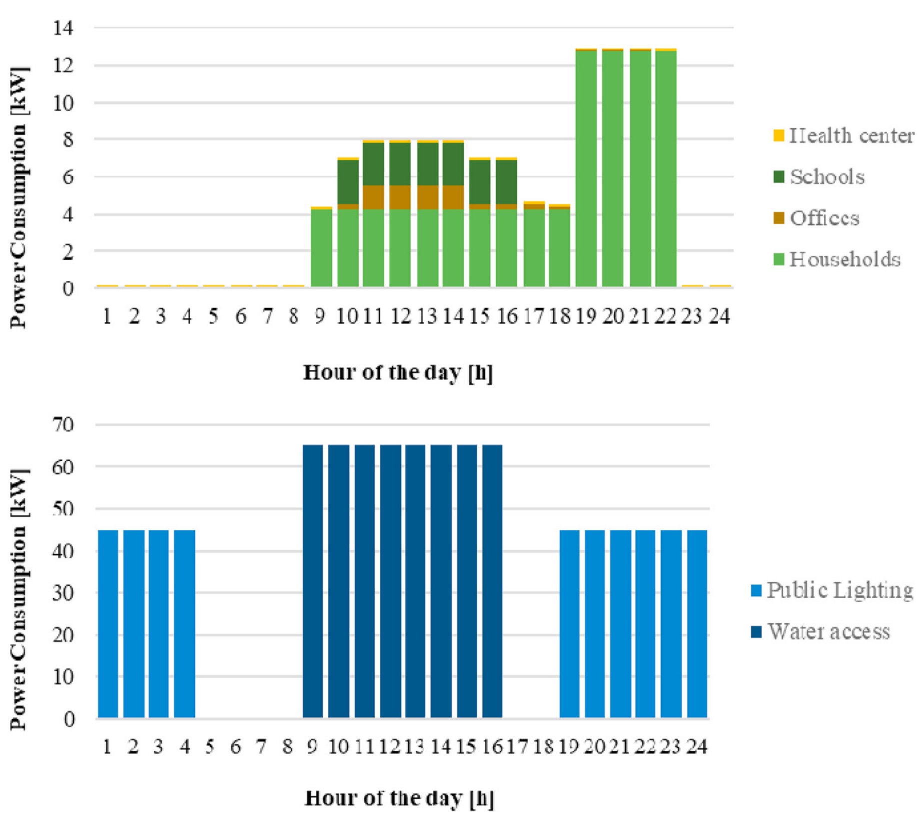

Fig. 3. Average daily electricity consumption profile of Mantapala refugee camp.

a higher electrification rate and access to renewable mini-grids, electricity demand may increase.

Alongside it, one could also assume that the buildings' electricity demand (households, schools, etc.) could be assured by home-solar-systems, as they are now in some cases. However, the precarity and temporary nature of the households' construction do not allow for such installations, neither do comply with the long-term planning vision of providing electricity for the whole community as a hybrid renewable mini-grid. Stable electricity access in refugee households is of extreme social importance, as being able to charge a mobile phone and have access to light after sunset will further allow for refugees' self-initiatives like creating their own business and/or furthering their education, thereupon decreasing their dependency on humanitarian aid and thus their vulnerable situation.

4.1.2. Cooking needs

Regarding cooking needs, the methodology developed in Section 2.1.2 was applied. Therefore, considering the data collected for Mantapala camp, we get the daily useful cooking energy need and associated emissions, respectively from Equations (2) and (3), as presented in Table 8.

4.1.3. Biogas production potential

To calculate the biogas production potential of Mantapala Refugee camp, we examined the potential of the human waste collection at the camp (applying Equation (5)), resulting in a higher value when compared to the biogas need to fulfil cooking needs (Equation (4)).

Biogas could be either collected just for cooking purposes, improving cooking efficiencies, or equally, to be fed (partially or 100%) into a biogas generator for the general electricity supply of the camp. Thus, Table 9 presents the biogas' production potential (E_{biogas}), biogas' cooking needs ($E_{f,biogas}$), and biogas availability for electricity production ($E_{biogas,electricity}$) for the Mantapala refugee camp.

4.2. Interviews

From the interviews, it was possible to identify several technical and economic challenges in the implementation of alternative power production systems.

Looking at the technical side, the main barriers identified were:

- integration and control of the different components;
- minimization of operations and maintenance procedures; and
- reduction of cost of the multiple technologies.

These barriers were pointed as critical areas on the path to improving the acceptance of these systems, and where research is still very much required. However, all interviewees agreed that, although challenging, none of these barriers required any breakthrough technology to be overcome.

The economic concerns identified during the interviews were mostly risk-taking related. Humanitarian operations inside refugee camps are characterized by uncertainty, from the moment it is settled until it is eventually phased out. The difficulty in predicting the dimension and duration of the refugee camp complicates budget management and discourages organizations to make long-term investments. Additionally, as humanitarian organizations almost exclusively depend on donor funding, their annual budget may fluctuate throughout the lifetime of a refugee camp. In fact, the budget tends to progressively decrease due to donor fatigue, forcing organizations to adopt a short-term planning strategy. According to the interviews' findings, "*Short-term budgeting inhibits the implementation of long-term strategies for technology introduction, such as solar panels and windmills, which are economically beneficial in the long term (…) the biannual budgets are incompatible with creating economically viable and, at the same time, environmentally sustainable energy solutions for a population. The increased use of long-term budgets presents opportunities both for businesses,* funding *institutions and humanitarian agencies*". However, long-term investment is still extremely rare in this sector, and can only be achieved by few humanitarian organizations. Consequently, energy is commonly not prioritized when compared with other utilities such as food provision or medical equipment, making it even harder to attract budgetary resource towards improved power production technologies.

Although there is a consensus regarding the economical attractiveness of hybrid renewable energy systems for humanitarian purposes, there are still barriers to overcome. Firstly, as observed during the interviews, humanitarian agencies are reluctant to be associated with the private sector, as they seem to "*speak different languages (…) due to their extremely distinct motivations*". For this reason, negotiations between these two distinct stakeholders are usually complex and time-consuming, making it difficult to create cooperative projects, especially on a long-term basis. Secondly and most importantly, uncertainty regarding sales, manufacturing and stock management makes this market unattractive for traditional business models. One of the interviewees, CEO of a supplier of stand-alone power supply systems for rural communities in developing countries, stated that "*there are already companies advancing towards the commercialization of these solutions, especially for the electrification of remote areas. (…) However, very few of these systems are design to be rapidly deployed to an emergency scenario*". Consequently, R&D investment in power supply systems for these scenarios is scarce, as companies find this market too risky.

4.3. Tecno-economic analysis

Based on the data and assumptions described in the previous sections, a techno-economic optimization of a hybrid renewable energy system for Mantapala refugee camp, is presented and discussed below.

First, a simulation was performed considering that the available renewable technologies would consist of solar PV and wind, as well as batteries. On a second stage, a biogas generator was also considered with the total potential (E_{biogas}) or, alternatively, a partial potential ($E_{biogas,electricity}$), considering that the remaining share would be used for cooking. All solutions were compared against a baseline scenario of diesel generation and presented in Table 10. These cases represent the best technological configurations in terms of economical, energy and environmental indicators, such as initial investment, cost of energy, payback time, renewable fraction,

Table 8
Cooking energy needs and emissions for Mantapala Refugee Camp.

$E_{u,cooking}$ [MJ/day]	144,251
$CO_{2\,emi,cooking}$ [tonCO$_2$/day]	104

Table 9
Biogas needs and production potential for Mantapala Refugee camp.

$E_{f,\ biogas}[MJ/day]$	248,709
$E_{biogas}[MJ/day]$	578,708
$E_{biogas,electricity}[MJ/day]$	329,998

Table 10
Simulation results for different system configurations obtained from HOMER (Project lifetime = 10 years; diesel cost = 0.87 $/l; B stands for scenarios considering the use of biogas).

Technology	System Configuration						Power Production									Economic				
	PV	Wind	Biogas	Diesel	Battery	Converter	PV	Wind	Biogas	Diesel	Total	Renew. Fraction	Electri-city excess	CO_2 Savings	Fuel savings	Initial Invest.	NPC	COE	Pay-back	Savings
Unit	kW	kW	kW	kW	#	kW	kWh/yr	kWh/yr	kWh/yr	kWh/yr	kWh/yr	%	%	%	L	$	$	$/kWh	yr	$
Baseline scenario	0	0	0	120	0	0	0	0	0	459,874	459,874	0.0	**13.5**	0.0	0.0	**72,000**	1,824,584	0.459	–	–
System 1	180	0	0	120	195	106	317,491	0	0	201,121	518,612	49.4	18.7	55.6	**79,946**	396,421	**1,108,854**	**0.279**	**3.6**	**715,730**
System 2	208	10	0	120	152	116	366,599	3669	0	200,714	570,981	**49.5**	26.4	55.5	79,885	433,549	1,130,651	0.284	4.1	693,933
System 3	329	0	0	120	0	104	578,324	0	0	226,481	804,805	43.1	47.9	48.7	70,089	456,752	1,246,502	0.313	5.1	578,082
System 4	325	10	0	120	0	121	572,098	3669	0	222,742	798,509	44.0	47.5	49.6	71,359	486,218	1,254,072	0.315	5.3	570,512
System 1B	185	0	60	0	166	102	325,596	0	201,417	0	527,012	**100.0**	20.1	**99.9**	**143,902**	404,696	**738,942**	**0.186**	**2.4**	**1,085,642**
System 2B	183	0	60	120	168	102	321,858	0	201,668	108	523,635	100.0	**19.6**	99.8	143,868	475,275	751,547	0.189	3.0	1,073,037
System 3B	206	10	60	0	148	112	362,256	3669	197,641	0	563,565	100.0	25.3	99.9	143,902	444,721	758,746	0.191	2.7	1,065,838
System 4B	206	10	60	120	148	112	362,256	3669	197,488	187	563,599	100.0	25.3	99.8	143,845	516,721	771,103	0.194	3.4	1,053,481

and CO_2 emissions, so as to achieve the highest economic savings.

Looking at the first set of solutions without biogas, we see that S1 presents the highest savings with a PV-battery-diesel mini-grid, with the majority of the energy (61% coming from solar PV) being backed up by diesel generators. S1 presents a renewable fraction of 49% and 55% CO_2 emission savings when compared to the baseline scenario. Although with a considerable higher initial investment (five times higher), it accomplishes up to 716 k$ savings in the lifetime of the project (10 years) with a COE of 0.279 $/kWh and a payback time of 3.6 years. When looking at a solution using wind energy (S2), we observe that only less than 1% derives solely from wind energy, resulting in an almost irrelevant advantage in environmental terms and costlier in NPC and payback time. The low benefit of this scenario is mainly due to the wind energy's low production (average wind speed lower than 5 m/s), when compared to the solar PV's high potential, as presented in the Annex.

However, when considering the possibility of using biogas produced from human waste collection and treatment, the results are even more appealing. For both simulations using the different biogas potentials, results were the same in pointing out that the most electricity-economical production using biogas had already been achieved using partial biogas potential.

With a mini-grid configuration of PV-biogas-battery, S1B delivers around 63% from solar PV and 39% from biogas, providing the flexibility and backup given previously by diesel. In this case, the initial investment is also high, but the payback time is reduced to 2.4 years with a COE of just 0.186 $/kWh, presenting savings of up to 1 M$ in 10 years. The CO_2 emissions are reduced to zero, as the mini-grid runs exclusively on renewable energy.

Nonetheless, the level of uncertainty is most often higher in humanitarian operations, as the complex political, social and economic scenario can unexpectedly affect some of the project's variables. Acknowledging that refugee camps face multiple resilience challenges, the selected systems need to be both reliable and the most diversified in electricity-mix. Thus, in terms of resilience, system S4B consisting of a PV-Wind-biogas-battery-diesel mini-grid presents the best configuration: more capable of facing potential shortages of diesel or renewable sources, but with a still competitive COE of 0.194 $/kWh and a payback time of 3.4 years.

Although the initial investment required for hybrid renewable energy systems may be a setback in the adoption of this solution, when considering a long-term basis the baseline scenario is the worse power supply system, both economically and environmentally. As seen, mini-grid solutions with biogas generators seem to be a good option for increasing sustainability in refugee camps. Besides improving human waste treatment, one of the main challenges in refugee camps, it can also play an important role as local job creation for the community. Nevertheless, in our simulations the biogas resource was assumed to be free of charge. Therefore, choosing the ideal power supply system would depend on the specifications of each individual refugee camp, as well as the resources, goals and motivations of the humanitarian organizations on-site.

5. Discussion

Results from the previous section prove that replacing diesel generators with more resilient, efficient, and sustainable power production systems in refugee camps, is not only possible but also economically attractive. Technically, the development of such systems seems apparently straightforward. In fact, all technologies considered in the present analysis are already in matured phase, being widely used for multiple applications. Nonetheless, there are still both technical and non-technical challenges that must be overcome to increase the interest in these solutions.

Cost is a key factor in the adoption of any product or service. Results indicate that biogas generators and batteries present the largest initial investment from all the components under analysis. However, even with high acquisition costs, these two technologies are technically and economically viable for stand-alone HRES. Future cost reductions would result in lower initial investments and provide additional cost-savings for humanitarian organizations. Likewise, paybacks reported for different stand-alone HRES are of less than five years, which may prove attractive enough for private investment, introducing the possibility of alternative business models.

In terms of COE, our results report 0.186 $/kWh (PV-biogas-batteries) to 0.315 $/kWh (PV-wind-diesel) which are lower, albeit in line with the literature reviewed in Section 1, where a COE of 0.275 $/kWh was found for PV-wind-biogas-diesel-batteries (Salehin et al., 2011), 0.35 $/kWh for PV-wind-batteries-diesel (Chowdhury et al., 2020) and 0.4 $/kWh for PV-wind-dual generator diesel/biogas (Fuso Nerini et al., 2015).

On the other hand, the promotion of an enhanced cooking solution using biogas shows itself as another important step towards the promotion of energy efficiency in refugee camps, allied with the mitigation of indoor air pollution and decrease of potential health impacts.

Although stand-alone HRES and efficient cooking solutions have sufficient technical readiness level to be implemented in refugee camps, it is still necessary to persuade humanitarian organizations to adopt these solutions (Nielsen et al., 2013). According to the interviews, several barriers must be relieved within the humanitarian sector to allow for the successful adoption of the systems here proposed. The first issue to be tackled is the poor long-term planning cooperation between organizations, regarding power supply during humanitarian operations in refugee camps. It was also possible to conclude, that humanitarian organizations do not always have the proper expertise to perform the required technical assessments and question the current energy management practices. Consequently, more traditional technologies, with lower up-front costs, end up being implemented, such as diesel generators. Likewise, and contrary to other operational areas of humanitarian relief, such as food, health and shelter, the absence of an energy-focused cluster presents an additional challenge to the adoption of improved power supply solutions (The United Nations Office for the Coordination of Humanitarian Affairs, 2013). The lack of such an entity leads to an inefficient and environmentally harmful management of resources, as the adoption of individual solutions by each separate organization is common practice during humanitarian relief operations - even if globally more expensive and less efficient than a centralized power production solution. In such scenarios, the adoption of most alternative power production systems here proposed is extremely unlikely, since these can only be economically viable with the existence of an on-site centralized power production consortium between different humanitarian organizations. On that account, governmental and non-governmental entities should be encouraged to discuss the creation of a multi-agency cluster, especially focused on preparing and managing the energy supply in refugee camps.

5.1. Advancing on the implementation and business model sustainability

Results from the previous section have demonstrated that significant cost savings can be achieved with the implementation of alternative hybrid renewable energy systems in refugee camps. But how significant are these savings when compared with the global humanitarian aid budget?

In Mantapala refugee camp, where the total funding requirements hit 8.8 million $/year (UNICEF, 2018), we can find that even with a renewable energy mini-grid, related savings reach a maximum of 108 k$/yr, which represents only around 1% of the total humanitarian budget required. This value is below the energy expenditure for Goudoubo (Burkina Faso) and Dadaab (Kenya), of 8% and 20% respectively. These numbers might not appear appealing enough to encourage humanitarian organizations to review their energy management practices and invest in alternative power supply systems. However, these cost savings can interest other external entities. In fact, improved power supply systems for refugee camps can be an opportunity for private companies. By providing a service that every humanitarian relief operation needs but that no current organization is prepared or even interested in dealing with, a sustainable business model can be created around the implementation of HRES in refugee camps. Results indicate that paybacks associated with these systems are similar to energy-related investments in other sectors (Punda et al., 2017; Vides-Prado et al., 2018). Thus, the cooperation between private and humanitarian sector can help eliminate some of the existing investment constraints commonly found in these scenarios - e.g. risk, lack of capital for investment and insufficient technical knowledge to install new types of power supply units.

Several business models can be designed to optimize this synergy, in accordance to the different needs and limitations of both stakeholders. Although finding the best business model for these systems is outside the scope of this work, a possible solution is presented in Fig. 4.

As in other energy-related systems' investments, a company could provide a fully installed power supply system to the refugee camp, without requiring any initial investment from the humanitarian organizations. From then on, organizations would have to make monthly payments throughout the lifetime of the system, reflecting their power consumption plus a renting fee for using the power production system. This fee could be defined in different ways, depending on the social and geographic characteristics of the refugee camp. Generally, the monthly payments should be lower than the current expenditure on diesel but also enough to allow revenues for the company, at the end of the contract.

With risk-taking as the main barrier for investment by the private sector, the deployment of HRES in refugee camps could truthfully benefit from the cooperation between governmental and private entities, working towards an innovative, sustainable and risk-minimizing business model. Indeed, the initial phase of any refugee camp is anything but stable: the different refugees' arrival rates to the camps vary from crisis to crisis, and most settlements only reach a stationary dimension after several months. Consequently, all major infrastructures and services found in refugee camps tend to follow the same trend. As the techno-economic viability of HRES highly depends on the proper quantification of the energy demand profile within the refugee camp, most systems here analysed are not indicated for the initial transient phase. However, these solutions might be implemented at a later stage when the refugee camp reaches its maturity stage, and the quantification of its energy needs is much more precise. This gap between the settlement of the camp and its mature stage, offers time for companies to develop tailor-made power supply solutions, based on the specifications of each refugee camp.

In the meanwhile, during the transient phase of the refugee camp, less risky solutions such as S1 (PV-batteries-diesel) might be adopted (Table 10). Although lower on cost savings, the dimension of this solution can easily be increased over time, which is extremely important, especially during the initial phase of the camp. On the other hand, S1 can easily be integrated into long-term solutions adding a biogas generator as seen for S1B. Thus, a

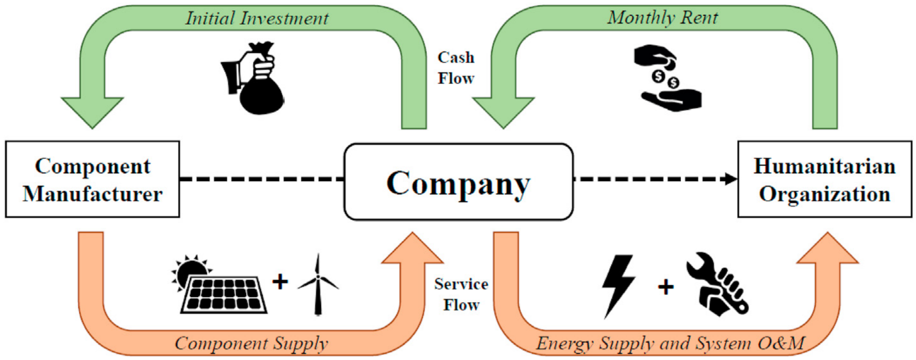

Fig. 4. *Graphical representation of a possible business model for the adoption hybrid renewable energy systems in the humanitarian sector.*

potential business model could offer long-term power supply contracts to humanitarian organizations, initially providing solutions similar to S1 and, later, upgrading to a S1B configuration. Therefore, due to their modularity, HRES can assure power supply from day one, while the system can be delivered, maintained and upgraded throughout the lifetime of the refugee camp. Nonetheless, a more detailed, individual analysis should be performed to identify the most adequate business model for each refugee camp.

6. Conclusions

In this work, the problems encountered by the absence of accurate information on energy demand needs and the inefficiency of power supply solutions found in refugee camps, were tackled by the development of a multi-disciplinary methodology that encompassed:

- the development of an energy demand modelling framework for Tier 0 refugee camps;
- the assessment of the technical and economic impacts of implementing alternative power supply systems in refugee camps in a safer and more sustainable way, by modelling the HRES with HOMER software; and
- the discussion of the challenges and barriers that may hinder the implementation of HRES in refugee camps, by interviewing relevant stakeholders in the humanitarian sector.

This methodology was applied to the Mantapala refugee camp, in Zambia.

Looking at the energy demand modelling, we understand that trying to input minimum standards of lighting and electricity access in refugee camps (meaning that Tier 0 situations would be improved to Tier 1) will result in a higher modelled electricity demand than the presently observed. Similarly, it is expected that higher electricity access, will increase energy demand by providing new opportunities for quality-of-life improvement (education, communication, business, etc.).

The results from the techno-economic analysis prove that diesel generators can be replaced with hybrid renewable energy minigrids, observing payback periods lower than five years and a COE of 0.315 \$/kWh. These decrease respectively to 2.4 years and a minimum COE of 0.186 \$/kWh, if a HRES with PV-biogas-batteries is considered, with the added advantage of enabling improved cooking conditions for daily use. This shift in power supply would also bring CO_2 emissions savings of 55%, and even up to 100% if a biogas generator is considered, since it would run exclusively on renewable energy, improving air quality along the way.

The viability of implementing such solutions in refugee camps was discussed, using information collected during a series of interviews performed with different stakeholders of this ecosystem. Although at a technical level HRES present high-readiness levels, problems such as uncertainty, lack of funding and difficulties related to risk-sharing collaborations were identified as possible constraints to their adoption in refugee camps. However, given the low impact of energy-related costs when compared with the overall humanitarian budget (around 1%), it is possible to conclude that the majority of these alternative power supply systems could be implemented in refugee camps in a near future, if some organizational changes in the humanitarian sector occur, namely developing tailored business models to HRES in refugee camps.

CRediT authorship contribution statement

Diana Neves: Conceptualization, Investigation, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing. **Patrícia Baptista:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **João M. Pires:** Conceptualization, Investigation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2021.126745.

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